

# PRITAM RANJAN KALITA

Project Assistant, Department of Electrical Engineering  
Indian Institute of Technology, Jodhpur, India

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## EDUCATION

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### M.Tech., Robotics Engineering

August 2024 – May 2026

*University of Petroleum & Energy Studies, Dehradun, India*

- Advisor: Dr. Abhinav Sharma
- Dissertation: “Uncertainty-Aware Trajectory Optimization and Robust Control for Grid-Based Mobile Robots”
- CGPA: 8.9/10

### B.Tech., Mechanical Engineering

August 2016 – May 2020

*Dibrugarh University Institute of Engineering & Technology, Dibrugarh, India*

- CGPA: 7.58/10

## EXPERIENCE & INTERNSHIPS

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### Project Assistant

July 2026 – Present

*Wearable Robotics & Control Laboratory,  
Indian Institute of Technology, Jodhpur, India*

*Supervisor: Dr. Saurav Kumar*

- Independently assembled and integrated a Powered Prosthetic Leg (Open Source Leg V2) from ground up using its individual mechanical, electrical and electronic components upto its full operational capacity.
- Implemented a time-based gait-cycle trajectory PD controller for coordinated knee and ankle joint actuation of the powered prosthetic leg simulating a human’s walk at 0.4m/s speed using open-sourced human gait trajectory data for control reference. 🎥

### Robotics and Automation Intern

January 2026 – June 2026

*PRD Robotech Solutions Pvt. Ltd., Noida, India*

*Supervisor: Mr. Prashant Kumar Dwivedi*

- Developed an obstacle detection and safety feature as part of a Robot Fleet Management Project for preventing collisions between differential-drive robots operating in a 3-way grid-map based warehouse environment using fused data from two 270° 2D LiDAR scans.
- Authored a 3-part technical training manual on the use of ROS 2 Diagnostics package for the company engineers, covering its C++ implementation for monitoring ROS 2 topics, services & actions.
- Created a custom Nav2 global planner plugin using the A\* path planning algorithm (without replanning) from scratch. Tested the algorithm on Gazebo Ignition and then implemented it on a differential-drive robot.

### Systems Engineer

April 2022 – August 2023

*Infosys Limited, Bhubaneswar, India*

- Worked on frontend development of customer websites using HTML, CSS, JavaScript, & React.

## PROJECTS

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### Uncertainty-Aware Trajectory Optimization and Robust Control for Grid-Based Mobile Robots 📄

January 2026 – May 2026

*M.Tech. Thesis Project, UPES Dehradun*

*Advisor: Dr. Abhinav Sharma, Mr. Prashant Kumar Dwivedi*

- Developed a Stochastic Nonlinear Model Predictive Control (SNMPC) framework on top of the existing Regulated Pure Pursuit (RPP) controller to optimize its lookahead distance parameter by jointly considering the robot’s path tracking error and localization uncertainty instead of relying solely on the velocity-based heuristic of the original RPP algorithm.
- The SNMPC-RPP controller uses a nonlinear path tracking error prediction model, path-relative error covariance propagation and chance constraints for determining an optimal value of the lookahead distance parameter while preserving the rest of the original RPP architecture.
- Validated the controller in ROS 2/Gazebo across multiple operating speeds, achieving reductions of over 40% in cross-track MAE and over 50% in RMSE compared with the baseline RPP controller at 1.5 m/s.

# Optimal Control of a Two-Wheeled Self-Balancing Robot Using LQR and Multi-Objective Genetic Algorithm

August 2025 – November 2025

*M.Tech. Third-Semester Major Project, UPES Dehradun*

*Advisor: Dr. Rahul Singhal*

- Derived a linearized state-space model for controlling the position and the chassis-tilt angle of a two-wheeled self-balancing (inverted-pendulum) robot, and implemented PID and LQR controllers using MATLAB/Simulink and Simscape Multibody.
- Implemented a Multi-Objective Genetic Algorithm (MOGA) to optimize the LQR weight matrices  $Q$  and  $R$  by simultaneously minimizing the Integral Time Absolute Error (ITAE) for both position and chassis-tilt angles.
- Compared the performance of the GA-optimized LQR with manually tuned LQR and PID controllers through MATLAB/Simulink simulations, demonstrating faster position convergence while maintaining better chassis-tilt angle stability.

## TECHNICAL SKILLS

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|----------------------------------------|-----------------------------------------------------------------------------------|
| <b>Programming Languages:</b>          | C++, Python, MATLAB                                                               |
| <b>Software &amp; Frameworks:</b>      | ROS 2, Nav2, Gazebo, Simulink, Simscape Multibody, SolidWorks                     |
| <b>Development &amp; System Tools:</b> | Linux, Git & GitHub, VS Code                                                      |
| <b>Robotics Hardware:</b>              | Arduino Uno, Raspberry Pi 5, CubeMars AK80-9 V3.0 Motor, AS5048B Magnetic Encoder |
| <b>Documentation:</b>                  | LaTeX, Markdown                                                                   |

## RESEARCH INTERESTS

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|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Robotics Fields:</b> | Medical Robotics, Lower Limb Powered Prostheses & Exoskeletons, Aerial Robotics, Space Robotics, Underwater Autonomous Vehicles |
| <b>Control Theory:</b>  | Model Predictive Control, Stochastic Control, Optimal Control, Non-Linear Control                                               |
| <b>Others:</b>          | Sensor Fusion, Computer Vision, Mapping, Localization, Applications of Learning Algorithms in Robotics                          |

## REFERENCES

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**Saurav Kumar, PhD**

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Department of Electrical Engineering  
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